Scenarios:

1. With phylogenetic model importation

Hypothesis: the isolation probability is much higher after March 2023, what FoI should we use?

- P(isolation | symp, aware) = 0.2 → 0.5
- FoI = ?

		P(Isolation / diagnosed, symptomatic)					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

Hypothesis: FoI back to the pre-August 2022 level, = 2.2, What should P(isolation | symp, aware) be?

- Fix FoI = 2.2
- P(isolation | symp, aware) = ?

		P(Isolation / diagnosed, symptomatic)					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

Hypothesis: Can behavior change in months where we see most of the importations prevent increase in cases?

- For week 4 – 16 (Apr-Jun) and week 26 – 34 (Sept-Oct), FOI = 0.7 (S11)

- For week 4 – 16 (Apr-Jun), FOI = 0.7 (S15)
- For week 26 – 34 (Sept-Oct), FOI = 0.7 (S16)
- FOI = 2.2 for the other time periods

Hypothesis: Can behavior change with the same number of months as in the previous hypothesis prevent increase in cases?

- Randomly select 22 weeks with FoI = 0.7 - full model period (S17)
- Randomly select 13 weeks with FoI = 0.7 - full model period (S18)
- Randomly select 9 weeks with FoI = 0.7 - full model period (S19)
- Randomly select 22 weeks with FoI = 0.7 – model period in 2023 (S25)
- Randomly select 13 weeks with FoI = 0.7 - model period in 2023 (S24)
- Randomly select 9 weeks with FoI = 0.7 - model period in 2023 (S23)
- FOI = 2.2 for the other time periods

Hypothesis: P(ISO) remains the same as the previous levels, what should the FOI be?

		P(Isolation | diagnosed, symptomatic)					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.45		S12			S2	
	2.2	S7	S6	S5	S4	S3	S14

		P(Isolation | diagnosed, symptomatic)					
		0	0.2	0.3	0.4	0.5	0.8
FOI	0.7		S13			S1	
	1.6		S20			S2	
	1.8		S21				

	2.0		S22				
	2.2	S7	S6	S5	S4	S3	S14

2. **Without phylogenetic model importation**

- FOI = 0.7, P(ISO) = 0.2, no importation **(S0)**
- FOI = 2.2, P(ISO) = 0.2, no importation **(S26)**
- FOI = 0.7, P(ISO) = 0.2, constant importation 5/10/15 every week **(S8, S9, S10)**

File names and meaning:

- **new_mpox2024_S#**: Baseline scenario S#.
- **0.5Xnew_mpox2024_S#**: Same as baseline scenarios but all specified 50% of phylogenetic importations.
- **2Xnew_mpox2024_S#:** Same as baseline scenarios but all specified doubling of phylogenetic importations.
- **import_undiag_mpox2024_S#:** Same as baseline scenarios but phylogenetic importations are imported as undiagnosed (as the default is importations are diagnosed).
- **import_asymp_mpox2024_S#:** Same as baseline scenarios but phylogenetic importations are imported as asymptomatic and so also undiagnosed (as the default is importations are symptomatic and diagnosed).
- **veX_mpox2024_S#:** Vaccine effectiveness wanes to X the baseline VE profile (e.g., ve0.25 = vaccine effectiveness wanes to 25% of baseline after a year of uptake) for scenario S#. Default setting X = 0.5.
- **2Xvax_mpox2024_S#:** Probability of vaccine uptake is 2 times of the baseline for scenario S#.
- **4Xvax_mpox2024_S#:** Probability of vaccine uptake is 4 times of the baseline for scenario S#.
- **10Xvax_mpox2024_S#:** Probability of vaccine uptake is 10 times of the baseline for scenario S#.